

Forecasting Volatility with Machine Learning and Rough Volatility: Example from the Crypto-Winter

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Abstract. We extend the application and test the performance of a recently introduced volatility prediction framework encompassing LSTM and rough volatility. Our asset class of interest is cryptocurrencies, at the beginning of the “crypto-winter” in 2022. We first show that to forecast volatility, a universal LSTM approach trained on a pool of assets outperforms traditional models. We then consider a parsimonious parametric model based on rough volatility and Zumbach effect. We obtain similar prediction performances with only five parameters whose values are non-asset-dependent. Our findings provide further evidence on the universality of the mechanisms underlying the volatility formation process.

Keywords – Machine learning, LSTM, rough volatility, quadratic rough Heston, Zumbach effect, cryptocurrencies, Bitcoin

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1 Introduction

In [31], classical parametric models, non-parametric Long Short-Term Memory (LSTM), as well as a novel parsimonious rough volatility-based predictor are used to predict stocks' volatility. The authors are successful in establishing some universal mechanism in the volatility formation process, and the rough volatility-based framework of [31] is able to forecast volatility better than traditional methods. In this paper, we are thus interested in testing the applicability and robustness of this approach by adapting the volatility forecasting framework to a different class of asset. Here we choose cryptocurrencies which is a much newer and younger market. Serving as a companion paper to [31], we aim at answering two questions in this work:

- Whether the devices used in the original paper predict volatility well for cryptocurrencies at the beginning of the so-called “crypto-winter”.
- Whether the volatilities have the same properties as observed for stocks, such as the universality of their formation process across coins during this period.

Bitcoin, proposed in [28], is a decentralized payment system where transactions are stored with blockchain technology, leveraging cryptographic algorithms. Many similar systems were introduced and formed a completely new asset class known as cryptocurrencies. Such electronic currencies offer many unique advantages over traditional fiat money. Since conception, cryptocurrencies have been a popular research topic in the academic community and have also received a lot of interest from financial institutions and the general public alike [2, 6, 10, 15, 19, 26]. The recent work [32] provided evidence that Bitcoin's volatility is rough, further justifying the inclusion of the rough volatility model in our work. The rapid increase in the num-

ber of listed coins on major cryptocurrency exchanges in recent years makes it more feasible to apply machine learning models such as the LSTM, and to demonstrate universality features.

The year 2022 corresponds to a significant price and volume drop in cryptocurrencies compared to the peak in late 2021, or the beginning of a “crypto-winter”^{[1][2][3][4]}. For example, we display in Figure 1 the prices and volumes from the two largest coins Bitcoin (BTC) and Ethereum (ETH).

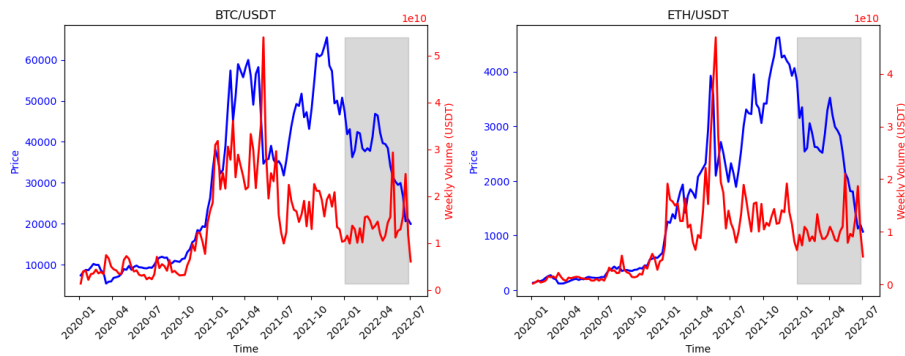


Figure 1: Price and weekly USDT trading volume of BTC and ETH, with the period Jan to Jun 2022 shaded.

There have been a few studies focusing on using machine learning to predict cryptocurrency volatility. In [11], deep neural networks are used to model the volatilities of the three most popular cryptocurrencies. The papers [24] and [1] consider a hybrid

¹Mccrank, J. (2022, February 14). Wall St Week Ahead Crypto investors face more uncertainty after rocky start to 2022. Reuters. <https://www.reuters.com/business/finance/wall-st-week-ahead-crypto-investors-face-more-uncertainty-after-rocky-start-2022-2022-02-11/>

²Harrison, E. (2022, May 10). The crypto winter is here. Bloomberg.com. <https://www.bloomberg.com/news/newsletters/2022-05-10/the-crypto-winter-is-here>

³Howcroft, E. (2022, June 13). Cryptocurrency market value slumps under \$1 trillion. Reuters. <https://www.reuters.com/business/finance/cryptocurrency-market-value-slumps-under-1-trillion-2022-06-13/>

⁴CoinGecko. (2023). 2022 Annual Crypto Industry Report. CoinGecko. <https://www.coingecko.com/research/publications/2022-annual-crypto-report>

approach combining neural networks and GARCH models to model cryptocurrency volatility, with the former using only Bitcoin and the latter using 27 coins. Compared with existing literature, we use a much larger set of 213 cryptocurrencies which enables us to show the universality of the volatility formation process across the asset class.

The rough volatility paradigm allows us to generate stylized facts of realized volatility [5, 16]. It is also effective in pricing and hedging derivatives [4, 12, 13, 14]. With the simplest version of this approach, namely the rough fractional stochastic volatility (RFSV) model, volatility forecasts can be made with a simple formula that requires essentially one parameter, namely the Hurst parameter. The Hurst parameter is systematically around 0.1 for all different asset classes [5, 16, 33], giving a quasi-universal volatility model and suggesting universality in the volatility formation process across assets. In this paper, we also estimate the Hurst parameter for many cryptocurrencies and demonstrate the same observation.

In [31], the authors presume that the universal volatility formation mechanism involves past returns. They extend the RFSV approach incorporating strong Zumbach effect [34] which corresponds to the feedback effect of price trends on volatility. To do so they incorporate a quadratic rough Heston (QRH) component to the RFSV. We recall the QRH model is particularly popular as it is able to reproduce market observations of SPX and VIX options [17, 30] thanks to its rough nature and the Zumbach effect.

We focus on the problem of forecasting the daily volatility of cryptocurrencies one day ahead. In particular, one of our goals is to understand the universal aspects

of the volatility formation process. One element towards this will be to investigate whether non-asset-dependent, generic, forecasting devices can lead to optimal predictions of future volatilities. First, we show that the LSTM approach performs better than classical autoregressive models, one of which is the heterogeneous autoregressive (HAR) model [8]. Then we use a combination of the RFSV and QRH for the parsimonious model candidate. We also conduct comparison tests to show that for this model, the universal version of the model performs very similarly to the one calibrated on each coin. This further confirms the universality of the volatility formation process from a parametric perspective.

We will follow the same structure as our companion paper [31]. In Section 2, we describe our data and model evaluation metrics. In Section 3, we describe both the parametric and non-parametric forecasting devices. We compare in Section 4 the performances of different forecasting models to provide evidence of a universal volatility formation mechanism for cryptocurrencies. Our approach also enables us to uncover asymmetry properties in the cryptocurrencies' volatilities in contrast to the stocks' volatilities. In Section 5, we aim to describe a universal mechanism for the volatility formation process with a parsimonious parametric model relating future volatility with past volatilities and returns. We conclude in Section 6.

2 Data and evaluation metrics

Our dataset contains 5-minute intraday prices of cryptocurrencies traded on the Binance exchange, with USDT, a popularly used stablecoin, as the base currency. Stablecoins are cryptocurrencies with price stabilization mechanisms, often in the form of pegging to some other assets such as traditional fiat money or other cryp-

tocurrencies [27]. In the case of USDT, it is pegged 1-to-1 with US dollars and can be considered a cryptocurrency version of USD⁵. The Binance exchange is one of the largest cryptocurrency exchanges by trading volume and offers easily obtainable market data, including price, volume, and number of trades in intervals as granular as 5 minutes. The data is directly obtained from Binance.com.

We select 213 coins after filtering out coins that are stablecoins, leveraged, or have low liquidity or insufficient data. The training set is chosen to be the two-year period from *2020-01-01* to *2021-12-31*, and the test set is a six-month period from *2022-01-01* to *2022-06-30*.

Due to the rapid increase in the number of cryptocurrencies in recent years and the issue of past data availability, we choose not to use training data from before 2020. The distribution of the number of available data points in the entire 2.5-year period for selected coins is given in Figure 2

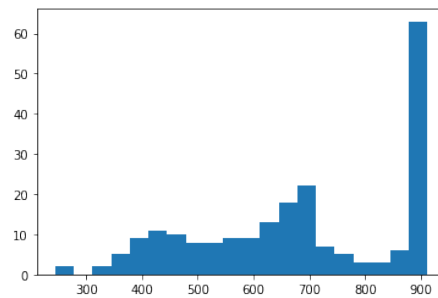


Figure 2: Distribution of the number of available data points for selected coins.

The daily realized volatility for day t , σ_t , the variable of interest, is computed

⁵More about USDT can be found at <https://tether.to/en/how-it-works>.

as

$$\sigma_t = \sqrt{\sum_i r_{t,i}^2},$$

where $r_{t,i}^2$ is the i^{th} 5-minute returns of the day. The return of day t is defined as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}},$$

where P_t is the closing price of day t . Since cryptocurrencies trade 24 hours per day every day, daily volatilities and returns are for 24-hour periods. The volatilities and returns are scaled and normalized respectively by the individual coin's statistics calculated on the training set.

For model evaluation, we choose to use the mean-squared-error or MSE, calculated as

$$MSE(\sigma, \hat{\sigma}) = \frac{1}{T} \sum_{t=1}^T (\hat{\sigma}_t - \sigma_t)^2,$$

where T is the number of trading days in the test set. According to [29], it is a robust and homogeneous metric for evaluating and comparing volatility models.

3 Forecasting devices

Here we introduce the parametric and non-parametric forecasting devices in this study.

3.1 Parametric models

First, we consider two auto-regressive type models. One is a classical autoregressive (AR) model, the other is a heterogeneous autoregressive (HAR) model [8]. Due

to the fact that cryptocurrencies are traded every day, for the HAR, the windows are chosen to be 7 and 30 instead of 5 and 22 in the literature on traditional markets. More precisely we define:

- AR_p

$$\hat{\sigma}_t = \alpha_0 + \sum_{j=1}^p \beta_j \sigma_{t-j}$$

p is chosen to be 7 or 30.

- HAR

$$\hat{\sigma}_t = \alpha_0 + \beta_1 \sigma_{t-1} + \beta_2 \sum_{j=1}^7 \sigma_{t-j} + \beta_3 \sum_{j=1}^{30} \sigma_{t-j}$$

These two models are fitted on individual coin data.

For the RFSV, the log-volatility is modeled as a fractional Brownian motion W_t^H :

$$d \log \sigma_t = \nu dW_t^H, \nu > 0$$

The Hurst parameter H and ν can be estimated using the method in [16].

As in [16, 31], the predictor for $H < \frac{1}{2}$ is given by:

$$\widehat{\log \sigma_t} = \frac{\cos(H\pi)}{\pi} \int_{-\infty}^{t-1} \frac{\log \sigma_s}{(t-s+1)(t-s)^{H+\frac{1}{2}}} ds,$$

$$\hat{\sigma}_t = c \exp(\widehat{\log \sigma_t}),$$

where $c = \exp(\frac{\Gamma(\frac{3}{2}-H)}{2\Gamma(H+\frac{1}{2})\Gamma(2-2H)}\nu^2)$. We will see in Section 5 that c is empirically very close to 1, and the impact of ν^2 on the forecast is marginal. H is calibrated using the pooled data set with all the coins.

3.2 Non-parametric forecasting with LSTM

The recurrent neural network (RNN) is suitable for sequential learning tasks. Among types of RNNs, the Long Short-Term Memory (LSTM) neural network, introduced in [21], is one of the most popular choices for predicting financial time series.

For each timestep t , the following computations are performed:

$$\begin{aligned} i_t^l &= s(W_{ii}^l x_t^l + b_{ii}^l + W_{hi}^l h_{t-1}^l + b_{hi}^l), \\ f_t^l &= s(W_{if}^l x_t^l + b_{if}^l + W_{hf}^l h_{t-1}^l + b_{hf}^l), \\ g_t^l &= \tanh(W_{ig}^l x_t^l + b_{ig}^l + W_{hg}^l h_{t-1}^l + b_{hg}^l), \\ o_t^l &= s(W_{io}^l x_t^l + b_{io}^l + W_{ho}^l h_{t-1}^l + b_{ho}^l), \\ c_t^l &= f_t^l * c_{t-1}^l + i_t^l * g_t^l, \\ h_t^l &= o_t^l * \tanh(c_t^l) \end{aligned}$$

where the superscript indexes the layer, c_t^l , h_t^l are the cell and hidden states, x_t^l equals the t -th element of the input sequence (σ_t) or (σ_t, r_t) when $l = 1$ or h_t^{l-1} for $l > 1$, $s(\cdot)$ is the sigmoid function, $*$ is the Hadamard product, $W_{\cdot}^l$ and $b_{\cdot}^l$ are parameters to learn from data. LSTMs use gates i , f , and o to capture long-range dependence in the sequential data and to avoid the vanishing gradient problem of classical RNNs [18, 21].

We use LSTMs with input length p , which are also chosen to be 7 or 30. The inputs can be volatilities or both volatilities and returns. We denote them by LSTM_{p,var} and LSTM_{p,ret} respectively. The LSTM consists of one LSTM layer and one dense output layer. The dimensions of h_t^1 for LSTM_{p,var} and LSTM_{p,ret} are

chosen to be 2 and 4 respectively. This number is tuned using a cross-validation split within the training data. We use the SiLU, introduced in [20], as the activation function.

Neural network fitting uses stochastic gradient descent and the results are noisy. Therefore we train 10 identically structured models, initialized with 10 different seeds. Lastly, we use the average of all their predictions as the final prediction. The optimizer we use is Adam, introduced in [23], with a learning rate of 1e-3. The LSTMs are trained using the pooled data set.

4 Towards universality in the volatility formation process

In this section, we compare different forecasting devices and show that we can achieve optimal performance with the universal ones. We also analyze the local sensitivities of the LSTMs and relate them to asymmetric volatility.

4.1 Parametric vs LSTM

In Figure 3, we have each model's MSE relative to HAR, grouped by coins. We can see that:

- AR_7 underperforms HAR.
- AR_{30} is on par with HAR.
- LSTMs outperform HAR in general.

- LSTMs with a longer window show significantly better than those with a shorter window.
- LSTMs with return inputs show significantly better performances than those without.
- RFSV outperforms HAR, is slightly better than LSTM_{7,var} and close to LSTM_{30,var}, but underperforms LSTM_{7,ret} and LSTM_{30,ret}.

These are largely consistent with the relative performances in [31].

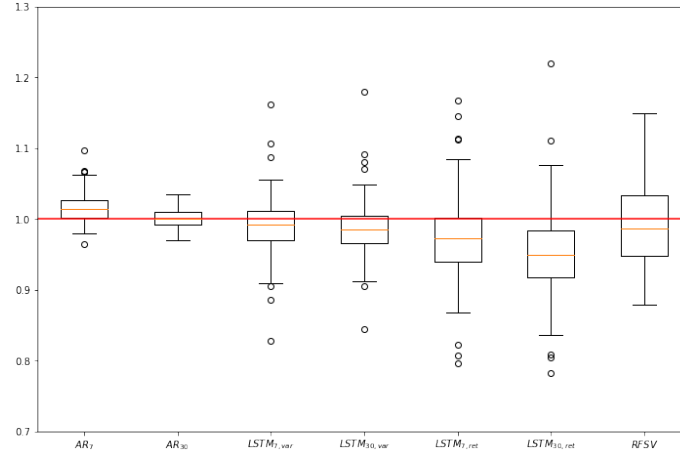


Figure 3: Distribution of out-of-sample MSE of each model relative to the HAR model.

4.2 Local sensitivities and asymmetric volatility

We define local sensitivities by:

$$\alpha_t(\tau) := \frac{\partial \hat{\sigma}_t}{\partial \sigma_{t-\tau}^2}, \quad \beta_t(\tau) := \frac{\partial \hat{\sigma}_t}{\partial r_{t-\tau}} \text{ for } \tau = 1, 2, \dots, 30.$$

Here we obtain the gradients of the predictions with respect to each of the inputs of LSTM_{30,ret}. For each coin, $\alpha(\tau)$ is given by the average of $\alpha_t(\tau)$, $\beta(\tau)$ the average

of $\beta_t(\tau)$. We then plot the average of these values across coins with a band of one standard deviation in Figure 4. The plot of $\alpha(\tau)$ looks very similar to the one for stocks in [31]. We observe that the values are significantly positive, have a similar magnitude, and decay with increasing τ .

For $\beta(\tau)$, the values are one order smaller than $\alpha(\tau)$, also decay in τ , and are positive for cryptocurrencies. In [31], $\beta(\tau)$ are observed to be negative for stocks. In both cases, the LSTMs captured a significant effect of returns on next-day volatilities, but the different signs of β for the two asset classes are a curious observation. If we use volatility as an indicator of trading activity or even the popularity of an asset, we might hypothesize that cryptocurrency traders react to positive returns more than to negative returns, while stock traders do the opposite. By extension, we can also speculate that the typologies of market participants are different between the two asset classes.

It has been previously observed that for cryptocurrencies, positive price shocks increase the volatility more than negative shocks do [3, 7, 22]. This phenomenon is known as inverted asymmetric volatility. In traditional markets, such as the stock market, the observed asymmetry is reversed, *i.e.* negative price shocks increase volatility more than positive shocks. Such an observation has often been attributed to the “fear of missing out” (FOMO) of uninformed cryptocurrency traders [9], and the existence of “pump and dump” schemes [25]. A common conclusion in these previous works is that the observation comes from the difference in behaviour and trading activity by uninformed and informed traders dominating the cryptocurrency and stock market respectively.

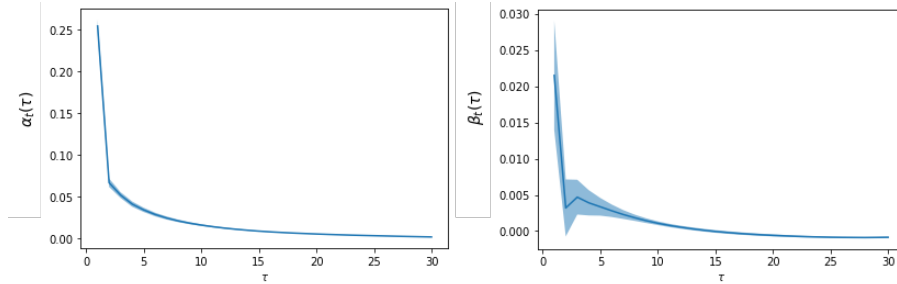


Figure 4: Average $\alpha_t(\tau)$ and $\beta_t(\tau)$ across coins and a one standard deviation band.

A nice byproduct of our methodology is a novel way to capture this inverted asymmetry in cryptocurrencies with the LSTM. We have also provided supportive evidence of this phenomenon across a much wider selection of cryptocurrencies than in previous studies of asymmetric volatility, confirming this observation on a large scale. Additionally, in tandem with [31], in which an asymmetry is also observed using a similar method, we have strong evidence from large baskets of assets from both asset classes that the asymmetry in cryptocurrencies is opposite from that in the stock market.

Figure 5 plots the model prediction $\hat{\sigma}$ and gradients $\alpha_t(\tau)$, $\beta_t(\tau)$ against σ_{t-1}^2 and r_{t-1} for all datapoints, without grouping by coin.

4.3 Comparing universal and non-universal LSTMs

4.3.1 Universality across different market capitalisation levels

A new LSTM_{30,ret} is trained on the top 50 coins by market capitalization (on 2021-12-31, the last day of the training period) only and compared with the universal one on the test period. The relative performance is shown in Figure 6. There is no degradation of performance by the much smaller set of coins used in training,

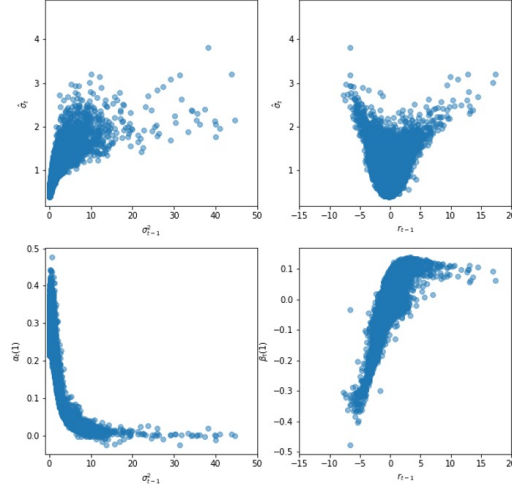


Figure 5: Top left: $\hat{\sigma}$ against σ_{t-1}^2 , top right: $\hat{\sigma}$ against r_{t-1} , bottom left: $\alpha_t(1)$ against σ_{t-1}^2 , bottom right: $\beta_t(1)$ against r_{t-1} .

which suggests some universality of volatility formation across coins at different levels of market capitalization.

4.3.2 No asset dependent mechanisms

Next, the universal LSTM_{30,ret} is fine-tuned to each individual coin's data to study whether this would improve its performance on individual coins. Fine-tuning of neural networks is done by initiating a neural network's parameters with a trained one, then further training it with an often smaller dataset. In our case, for each available coin, we initialize neural networks with the universal LSTM_{30,ret}, then train it further with only that coin's training data.

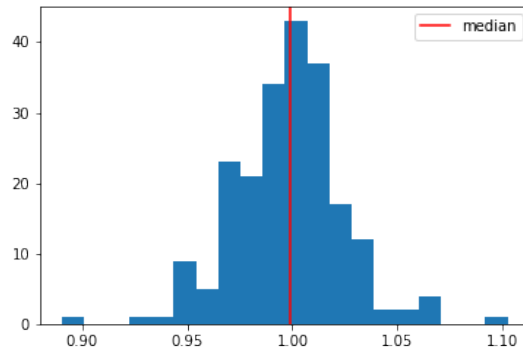


Figure 6: Distribution of out-of-sample MSE of LSTM_{30,ret} trained with the top 50 coins by market capitalization relative to the universal one.

If there were any asset-specific mechanisms that could be captured by the LSTM, we should see the fine-tuned LSTMs performing better. Here we allow all parameters to be changed during fine-tuning. The relative performance is shown in Figure 7. The fine-tuned models do not perform better. This supports that there are no asset-dependent mechanisms in volatility formation for the coins.

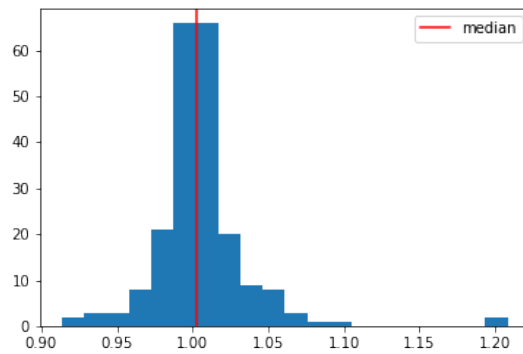


Figure 7: Distribution of out-of-sample MSE of LSTM_{30,ret} fine-tuned to each coin relative to the universal one.

5 Uncovering universal mechanisms in the volatility formation process

In this section, we calibrate the RFSV to each coin to confirm the universality of the volatility formation process. Then we incorporate the strong Zumbach effect into the rough-volatility based model by adding a quadratic rough Heston component. By comparing the combined model to the LSTM and RFSV, we obtain a promising candidate for the description of a universal volatility formation process.

5.1 RFSV

For RFSV, we first estimate H and c for each coin, and the distributions are shown in Figures 8 and 9. H and c tend to be higher than the ones for stocks as found in [31]. The median of H and c are 0.103 and 1.06 respectively.

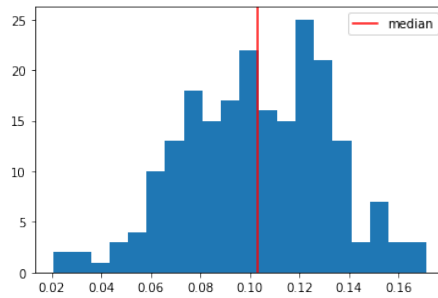


Figure 8: Distribution of H .

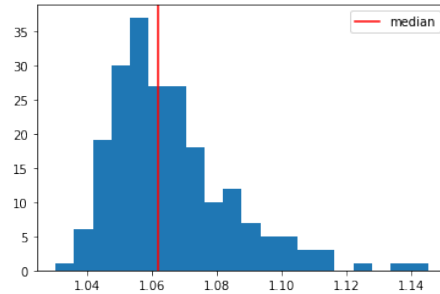


Figure 9: Distribution of c .

To once again check for universality from a parametric perspective, RFSVs are calibrated to each coin and compared with the universal RFSV that is calibrated to all coins. The relative performance is shown in Figure 10. The asset-specific RFSVs perform very similarly to the universal RFSV. Similar to fine-tuned LSTMs, we would expect improvements from the asset-specific RFSVs if there were any coin-specific mechanisms that could be captured by RFSV. Once again we obtain further evidence of a lack of asset-specific mechanisms from a parametric perspective.

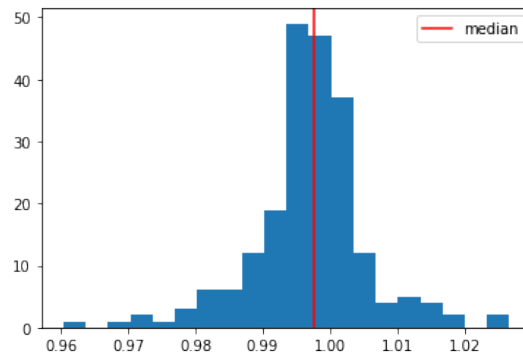


Figure 10: Distribution of out-of-sample MSE of RFSV calibrated to each coin relative to the universal one.

5.2 RFSV enhanced with QRH

Here we use the forecasting device based on the Zumbach effect from [30, 31] via a quadratic rough Heston component. The forecasting formula from the QRH model is given by

$$\begin{aligned}\hat{\sigma}_t^2 &= a(Z_{t-1} - b)^2 + c, \\ Z_t &= \sum_{i=1}^n c_i^d Z_{i,t}, \\ Z_{i,t} &= e^{-\gamma d_i} Z_{i,t-1} + r_t, \quad Z_{i,0} = z_{i,0}, \quad i = 1, \dots, n,\end{aligned}$$

where $a, b, c > 0$, and $(cd_i, \gamma d_i)_{i=1, \dots, n}$ are given by the same multi-factor approximation of the rough kernel function $K(t) := \frac{t^{H-1/2}}{\Gamma(H+1/2)}$ [30]. Given H and n , $(c_i^d, \gamma_i^d)_{i=1, 2, \dots, n}$ are not free parameters. The predictions are essentially moving averages of past realized volatilities and are not sensitive to the earliest inputs due to the exponential decay in weights. In practice after computing $Z_{i,t}$, one needs to discard the earliest samples without sufficient histories. By including the QRH, we have three more parameters a, b, c to calibrate which can be done by regressing σ_t^2 on (Z_{t-1}, Z_{t-1}^2) .

We will combine the predictions from RFSV and QRH linearly. The final prediction is given by $(1 - \lambda)\hat{\sigma}^{RFSV} + \lambda\hat{\sigma}^{QRH}$. The relative performances for different values for λ to RFSV only are shown in Figure [11] the relative performances to LSTM_{30,ret} are shown in Figure [12]

Incorporating past returns in the RFSV through the QRH component is a clear improvement. At $\lambda = 0.15$, the model performs best and is very similar

to $\text{LSTM}_{30,\text{ret}}$. This gives us a parsimonious parametric model candidate for a universal volatility mechanism.

We also note the larger optimal value for λ than that in [31] for stocks. Past returns play a larger part in the volatility formation in cryptocurrencies than in stocks.

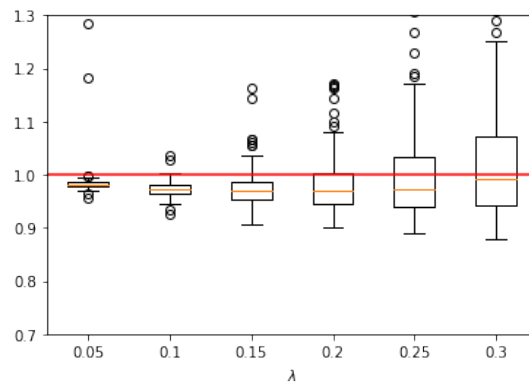


Figure 11: Distribution of out-of-sample MSE of $(1 - \lambda)\hat{\sigma}^{RFSV} + \lambda\hat{\sigma}^{QRH}$ relative to RFSV only (or $\lambda = 0$).

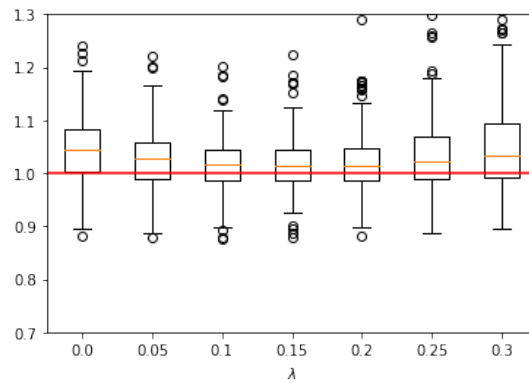


Figure 12: Distribution of out-of-sample MSE of $(1 - \lambda)\hat{\sigma}^{RFSV} + \lambda\hat{\sigma}^{QRH}$ relative to $\text{LSTM}_{30,\text{ret}}$.

6 Conclusion

In this work, we adapt the volatility forecasting framework of [31] and extend it to a different asset class. The framework and forecasting devices used in the original paper are applied to the daily volatilities of cryptocurrencies at the start of the “crypto-winter” crisis where prices and trading volume dropped. This is also the first study of cryptocurrency volatility in which a wide selection of coins is used. In general, we find the framework to work very well in producing relevant universal volatility forecasting devices, as well as evidence of universal mechanisms in the volatility formation process, as was found in stocks. The findings of the applied methods, regarding predicting volatility, are thus remarkably largely similar in the very different datasets and contexts.

Using classical auto-regressive models, LSTM and RFSV, we show that the volatility during this period can be modeled and predicted well with the latter two. The non-parametric LSTM and parsimonious RFSV, like for US stocks, outperform all other traditional models used in industry and past literature. While constructing the RFSV forecaster, we estimate the Hurst parameters of the log volatilities for all the coins and showed that their volatilities are indeed “rough”. While comparing the fine-tuned LSTM and RFSV to their universal counterparts, we observe a universality of this process across the wide selection of coins, which is also observed in stocks.

Next, we use the combination of RFSV and QRH to obtain a parsimonious model. We then find that the RFSV+QRH model can perform as well as the LSTM with both volatility and returns as input. This suggests that the main features in the volatility formation for cryptocurrencies can be described by rough volatility boosted with a strong Zumbach effect. This model has also once again confirmed

the universality of the volatility formation process across coins. The obtained models and volatility characteristics can further our understanding of this young and evolving market. Remarkably, this device is found to also perform very well for stock volatility, suggesting a higher level of universality across asset classes.

In addition, the use of LSTM offers a new way to capture the effect known as asymmetric volatility. By analyzing the gradients of the volatility forecast with respect to past returns, we can capture and confirm the asymmetrical effect of past returns on volatility, and compare this effect with other asset classes. This effect can be further used to infer the behavior and informedness of the traders in these markets. Based on existing results, we can hypothesize cryptocurrency traders tend to react to positive shocks more and might be less informed compared to traditional market participants.

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